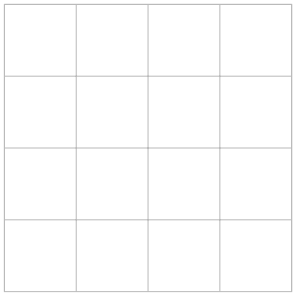
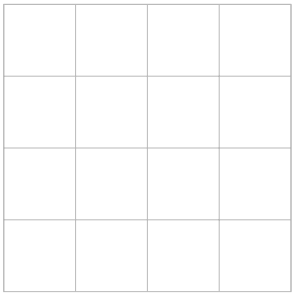
Keep all tiles flat and in one piece, joined along whole sides without overlaps. Count every side with no tile beside it, including sides around holes. Turns and flips count as the same shape.

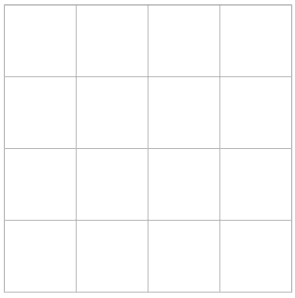
Problem 1: Make every different shape you can with four tiles. Draw each shape and count its boundary sides.



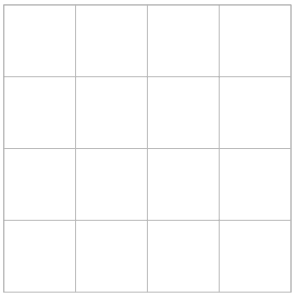
_____ sides



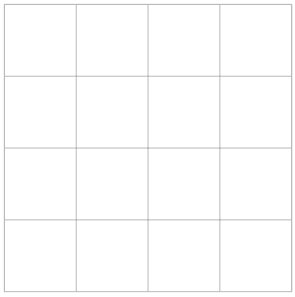
_____ sides



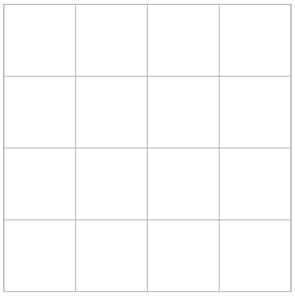
_____ sides



_____ sides

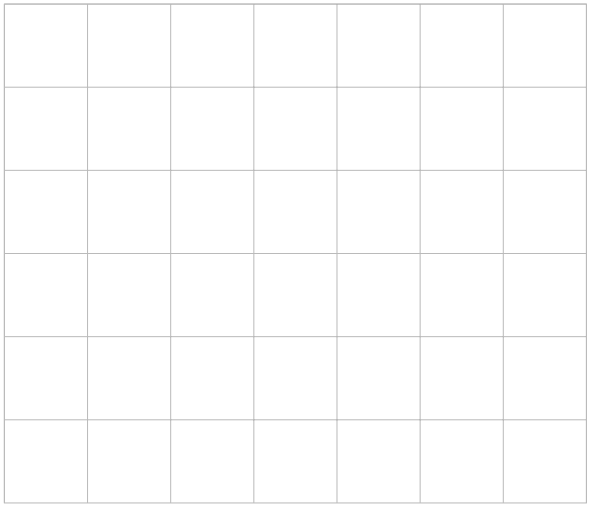


_____ sides



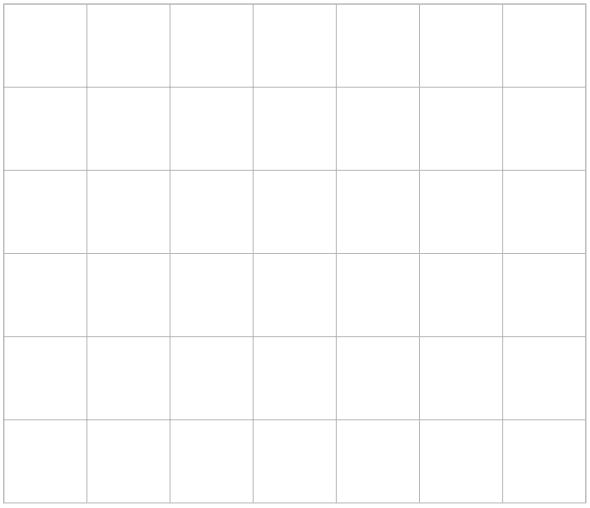
_____ sides

Problem 2: Make a shortest-boundary shape and a longest-boundary shape with 5 tiles. Do the same with 6 tiles.



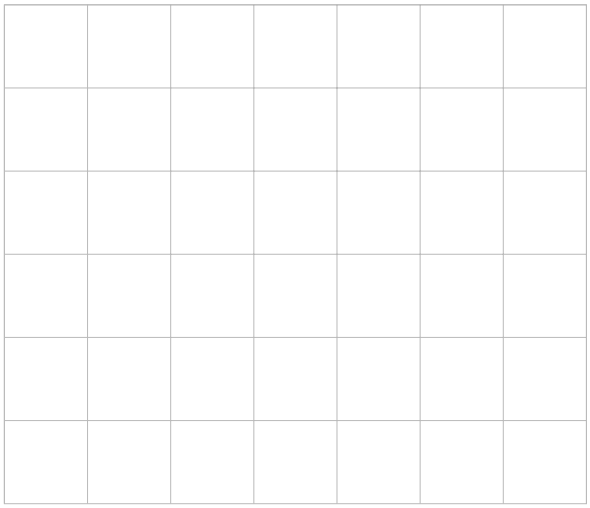
5 tiles; shortest

_____ sides



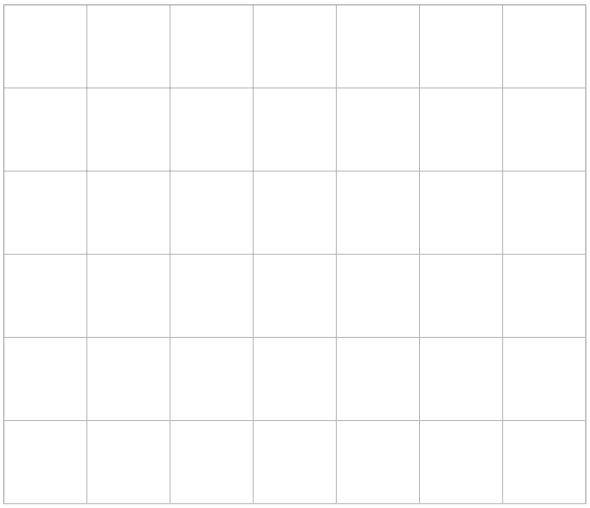
5 tiles; longest

_____ sides



6 tiles; shortest

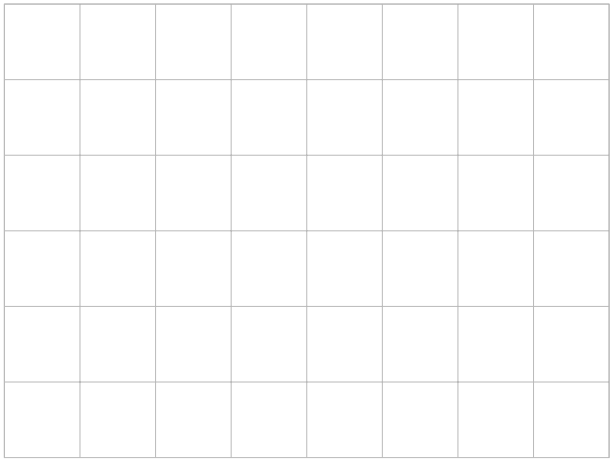
_____ sides



6 tiles; longest

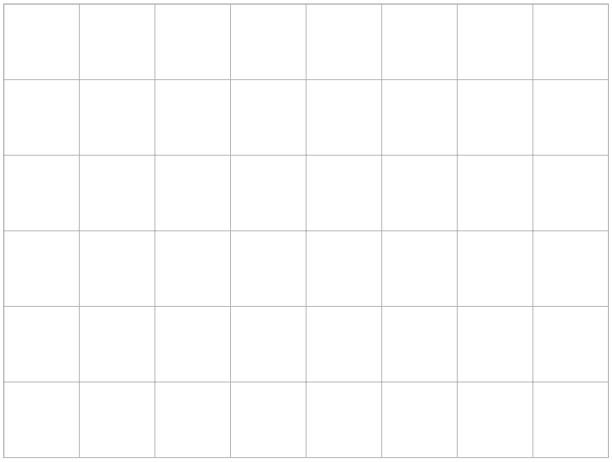
_____ sides

Problem 3: Make a shortest-boundary shape and a longest-boundary shape with 7 tiles. Do the same with 8 tiles.



7 tiles; shortest

_____ sides



7 tiles; longest

_____ sides



8 tiles; shortest

_____ sides



8 tiles; longest

_____ sides